VISION

To impart quality education in Information Technology and enable learners to cope with global challenges to build professional career benefitting the sustainable growth of an individual and the society at large.

MISSION

To propose state of the art educational environment equipped with cutting edge technology in the area of Information Technology.

To facilitate learners and faculties with every single opportunity of professional progression embedded in academic scenario itself which can cause enriched workforce contributing to the development of the nation.

To fulfill the noble cause of educating budding technocrats by accelerating the momentum of research and implementing innovative inputs in teaching-learning.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

1. To acquire knowledge of core area of engineering and suitable prerequisites through modern tools and techniques along with enhancing soft skills and continuing professional development.
2. To identify real life problems through proper investigation and to design and develop appropriate solution through systematic analysis which is economically feasible and in accordance with the need of industry, academia and society at large.
3. To exhibit the professional growth as an individual and a team as well; along with ethical responsibility and approach of lifelong learning.

PROGRAMME OUTCOMES (POS)

a. An ability to apply knowledge of computing, mathematics, science and engineering fundamentals in the information technology and diverse domain.

b. An ability to identify and analyze the problem to propose the appropriate solution.

c. An ability to design and develop information technology based solutions which satisfy the demand of the hour.

d. An ability to investigate and analyze the solution of complex problem with the help of current edge technology.

e. An ability to learn upcoming technologies and tools to fulfil the needs and necessities of the current trends.

f. An ability to explore the role of engineering in the progress of the society.

g. An ability to cultivate the sense of precautions for the environmental safety and other contemporary issues.

h. An ability to understand ethical responsibility of an individual to contribute towards the growth of nation.

i. An ability to work proficiently, individually and on teams, including multidisciplinary area, to achieve a common goal.

j. An ability to communicate effectively among the variety of audience.

k. An ability to develop the approach of lifelong learning for continuing professional development.

l. An ability to apply principles of management to achieve the targets within the limitations of available finance.

~INDEX~

Name:	Enrolment No:	Semester:

Sr. No	Name of the Experiment	Date	Page No.	Signature	Remarks
01	Study Python ecosystem for Machine learning: Python, SciPy, Scikit-learn.				
02	Study of pre-processing methods. Write a program to find following statistics from a given dataset. Mean, mode, median, variance, standard deviation, quartiles, interquartile range.				
03	Study and implement PCA in python.				
04	Study and implement simple linear regression.				
05	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.				
06	Write a program to implement the Naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.				
07	Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set. Compute the accuracy of the classifier, considering few test data sets.				
08	Write a program to implement SVM algorithm to classify the iris data set. Compute the accuracy of the classifier, considering few test data sets.				
09	Write a program to implement and check Sklearn's K-Fold, Shuffled K-fold, Repeated K-Fold and Leave-One-Out validation technique for appropriate classification algorithm and dataset.				
10	Write a program to implement k-Means clustering algorithm for a sample training data set stored as a .CSV file.				
11	Write a program to recognize handwritten digit using Artificial Neural Network.				
12	Study Weka toolkit for demonstration of regression, classification, and clustering models on it.				

G H PATEL COLLEGE OF ENGINEERINGAND TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

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SUBJECT NAME: MACHINE LEARNING

~*INDEX*~

Name:	Enrolment No:	Semester:

Sr. No	Name of the Experiment	Date	Page No.	Signature	Remarks
1	Assignment I				
2	Assignment II				
3	Mini Project				